Credit card overdue forecasting based on ensemble learning

Bing Han, August 2018

Voice of bank personnel: *"When banks are doing post loan management, they say that if they can know 5 days in advance that the customer will default, it will be a big profit. For banks, as long as they collect their debts one day earlier than other banks, their losses will be much smaller. Such applications will enable financial institutions to effectively control risks."*

Forecasting of Continuous Overdue

According to the length of overdue days, the credit card customer's overdue status can be divided into different time periods to measure the degree of customer default. The M1 period refers to the period with overdue days within 30 days, the M2 period refers to the period with overdue days greater than 30 days but less than 60 days, the M3 period refers to the period with overdue days greater than 60 days but less than 90 days, and so on. As the number of overdue days continues to increase, the likelihood of customers repaying all their debts decreases, ultimately leading to the inability to recover the debts and resulting in bad debts for the bank. Overdue payments exceeding M3 are considered serious defaults, and banks typically classify this group of people on a blacklist.

A ***roll rate*** is the percentage of delinquent accounts that continue through to the next 30-day period. In this study, we use a roll rate model to determine the bad samples, which are the positive samples in the classification model.The roll rate model calculates the probability of overdue status transitioning from existing time periods to larger or smaller time periods, as well as remaining unchanged. Based on this, the rolling rate of M0 (normal repayment users) transitioning to M1 (less than 30 days overdue), M1 transitioning to M2 (less than 60 days overdue), M2 transitioning to M3 (less than 90 days overdue), and other changes are obtained.

We sampled credit card overdue data with a window period of one year and calculated the roll rate results shown in Table 1.

Table 1. Roll rate result

Current period	Overdue status of next period (Probability)		
	Improved	Unchanged	Worse
M0		0.958	0.042
M1	0.538	0.38	0.082
M2	0.457	0.239	0.304
M3	0.41	0.204	0.386
M4	0.351	0.087	0.562
M5	0.283	0.078	0.639

From the table, it can be seen that the probability of customers in the M4 period rolling to a worse state is significantly higher than that of M3. Therefore, customers with overdue M4 and above are defined as "bad customers".

Before our modeling work, it is necessary to clarify the definition of continuous overdue behavior in this study. The predicted object specifically refers to users who have already experienced M1 overdue, have not yet repaid their debts, and have been continuously overdue until the M4 period. Define K as the number of credit card billing periods at the observation time point, D_K as the cardholder's overdue status for K-th period, $D_K \in \{M1, M2,..., MN\}$. The group we focus on is customers who have overdue in K-th period and are within 30 days of overdue, that is, the overdue status is D_K=M1. If the customer fails to repay the debt within the next 90 days, then after three billing cycles, the customer's overdue status D_{K+3} will be converted to M4, indicating a serious overdue. This type of customer belongs to the positive sample that the system should pay attention to, and is marked as 1. Otherwise, the customer is a negative sample and the class is marked as 0.

$$\text{Positive: } D_K = M1 \text{ AND } D_{K+3} = M4$$

$$\text{Negative: } D_K = M1 \text{ AND } D_{K+3} \neq M4$$

In addition, data filtering should also consider whether the cardholder is already on the blacklist of users at the issuing bank, and whether the cardholder's account meets a record of continuous payments for more than six months. If the cardholder is already on the bank's blacklist, it is obviously meaningless to make overdue forecasting. If the payment records of the cardholder’s account are less than six months, the conditions will not be met when calculating some historical indicators.

Considering the rapid growth of credit card business and the constantly changing customer structure in issuing bank, the observation period should not be set too long. This is because the increment of the credit card customer dataset will present new features over time, such as the average age of customers getting younger or their consumption characteristics becoming more diverse, which will affect the overall distribution of the data. Meanwhile, the observation period should not be set too short, otherwise many historical statistical indicators may not reflect the true trend due to insufficient data. With the considerations, we set the observation period to 6 months and the performance period to three months from the observation point. The effective range of the data is within one year.

Risk Dataset and Feature Engineering

We selected 24 features (Table 2) as independent variables to measure the overdue risk of credit card customers.These features cover multiple dimensions such as the general characteristics, economic status, credit status, and transaction preference of cardholders.

Table 2. Risk Dataset Feature List

Basic Information	**Previous Credit Status**
Gender (M: Male; F: Female)	Number of times entering the collection queue
Age	Number of overdue payments in the past 12 months
Education level	Maximum cumulative overdue times in the past 12 months
Annual income	Number of billing installments in the past 2 months
Industry Category Code (01: Education; 02: Culture; ...)	
	Installment and Withdrawal
Account Information	Cash withdrawals in the past month
Number of accounts	Single withdrawal amount in the past month
Account opening duration (days)	Historical highest consumption amount
Credit limit	
	Consumption Preferences
Current Billing Information	Jewelry consumption frequency
Overdue amount	Fitness consumption frequency
Last repayment amount	Furniture consumption frequency
The number of days from the last repayment to the billing date	Consumption frequency related to automobiles
Risk Preference Derivative Variables	
Historical maximum debt ratio (%)	
Average monthly limit use rate in the past two months (%)	

Account information not only reflects the customer's credit scale and loyalty, but also indirectly reflects the customer's daily financial situation. The current billing information measures the degree of M1 overdue. The derivative variables of historical liabilities and overdrafts in the past two months are used to reflect the customer's risk preference. A description of the customer's past credit status, such as the number of overdue payments in the past year, can determine whether the customer frequently experiences credit defaults.

The last four variables related to customer consumption preferences need to be separately explained. Generally speaking, for every Pay-By-Card, the bank records the type of merchant with whom the transaction occurred, namely MCC (Merchant Category Code). By using the MCC code, it is possible to determine what type of product the customer purchased. For example, MCC code 5812 corresponds to the industry code of UnionPay's catering merchants.

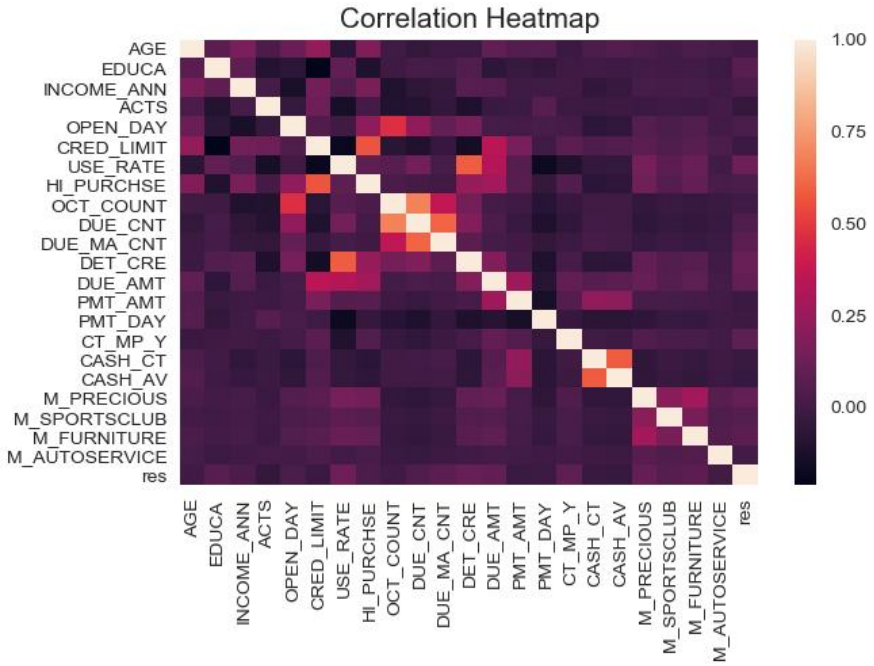
The MCC codes defined by UnionPay exceed 260, covering all aspects of daily consumption. However, in practice, having too many categorical variables can affect the complexity and interpretability of the model. One approach to reducing the range of categorical variables is to perform a ***one-hot encoding*** on the possible variable values. However, in decision tree models, this is not a good solution, especially when the dimensions of category features are high. The main reason is:

(1) Using one-hot encoding for features with higher dimensions results in less data in each category, leading to imbalanced dataset and minimal segmentation gain. In an unbalanced state, there is almost no difference between splitting and not splitting.

(2) One-hot encoding divides data into many scattered small spaces and make it difficult to obtain accurate statistical information which the learning of decision tree requires. As a result, the final learning effect will deteriorate.

Based on the above, we decided to select only four representative MCC categories, namely jewelry, fitness, furniture, and automobiles. As the business personnel said, if a customer frequently purchases jewelry products, it is likely that the customer has a higher income or a preference for luxury goods. If the customer has a record of purchasing furniture or cars, it can indicate that the customer has a house or a car. (As can be seen in the following text, the contribution of MCC type variables to the forecasting results is very limited.)

Figure 1. Correlation Heatmap



Feature engineering of datasets is an essential and crucial task. An important task in feature engineering is to test the correlation between feature values. As is well known, the higher the correlation between features and category labels, the better. However, a high correlation between features can lead to multicollinearity, which in turn leads to unstable forecasting results. One method to present the correlation between a large number of features is to calculate the Pearson coefficients of paired features and then plot them on a heatmap. It will base on the correlation coefficient matrix, in which every single element represents the correlation coefficient between two features.

Figure 1 shows the heatmap drawn from the credit card overdue risk dataset after removing the one-hot coded features. It can be seen that there is no significant strong correlation between any pairwise features, generally when we considered it to be greater than 0.8.

Modeling with Ensemble Learning Methods

For forecasting, our study selected two ensemble learning algorithms, AdaBoost and Random Forest, which performed exceptionally well in the field of machine learning at the time. Compared with other ***Boosting*** algorithms, AdaBoost shows a less

overfitting characteristic in experiments, and its upper bound of error rate tends to stabilize with increasing iteration times. The Random Forest using the ***Bagging*** strategy is also less prone to overfitting, and its generalization error will converge to an upper limit as the number of decision trees increases.

Compare the output of different forecasting methods and organize them to obtain Tables 3. Based on the above results, it can be concluded that both AdaBoost and Random Forest algorithms based on ensemble methods have a final precision of over 65%, which are superior to the other methods.

Table 3. Comparison of results from different forecasting methods

	Manual Evaluation	Decision Tree	Logistic Regression	AdaBoost	Random Forest
Precision	33.40%	60.70%	61.00%	**69.70%**	**67.00%**
Recall	25.80%	47.20%	48.30%	**53.20%**	**52.80%**
AUC	-	62.80%	63.40%	**74.30%**	**71.00%**

In order to achieve the best performance of the forecasting model on risk datasets, the system needs to continuously optimize the forecasting engine based on ensemble methods during the testing phase. The optimization method is to continuously try new algorithm parameters to further improve the performance of algorithms. Design the following testing process for this purpose:

STEP 1 - Load the raw data from the most recent observation period and perform data pre-processing;
STEP 2 - Using the model training function, select the reserve method to divide the training set and test set, with an allocation ratio of 7:3;
STEP 3 - Using ***random under-sampling (RUS)*** *on the training set to mitigate the impact of data imbalance;*
STEP 4 - Set the number of times the model is trained/tested repeatedly to 10 times;
STEP 5 - After the training is completed, calculate the average of the output results from all previous test sets as the final forecasting effect;
STEP 6 - Continuously adjust the model parameters and repeat the above steps.

Figure 2 shows the impact of different parameter values on AUC during AdaBoost training. The left figure shows the relationship between the number of iterations and model performance. In the early stages of training, as the number of iterations increases, the training effect of the model improves by leaps and bounds. As the number of iterations increases, there is no significant fluctuation in performance after more than 80 iterations, but there is still no obvious overfitting phenomenon, which fully reflects the characteristic of AdaBoost not being prone to overfitting. The figure on the right shows the different AUC obtained with different learning rates at an iteration of 80. As shown in the figure, setting ***learning_rate*** at 0.5~0.6 can obtain the maximum AUC. Setting ***learning_rate*** too low will obviously cause the convergence speed of the model to be slow, and after 80 iterations, it is impossible to learn good results; Setting ***learning_rate*** too high will result in missing the optimal solution.

Figure 2. AdaBoost parameter optimization process

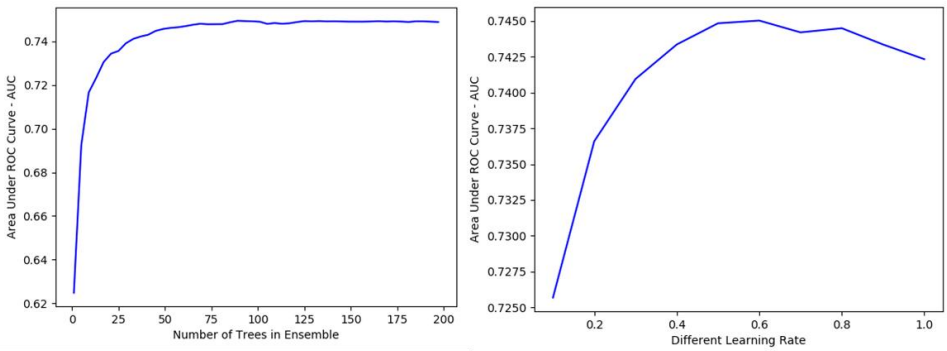
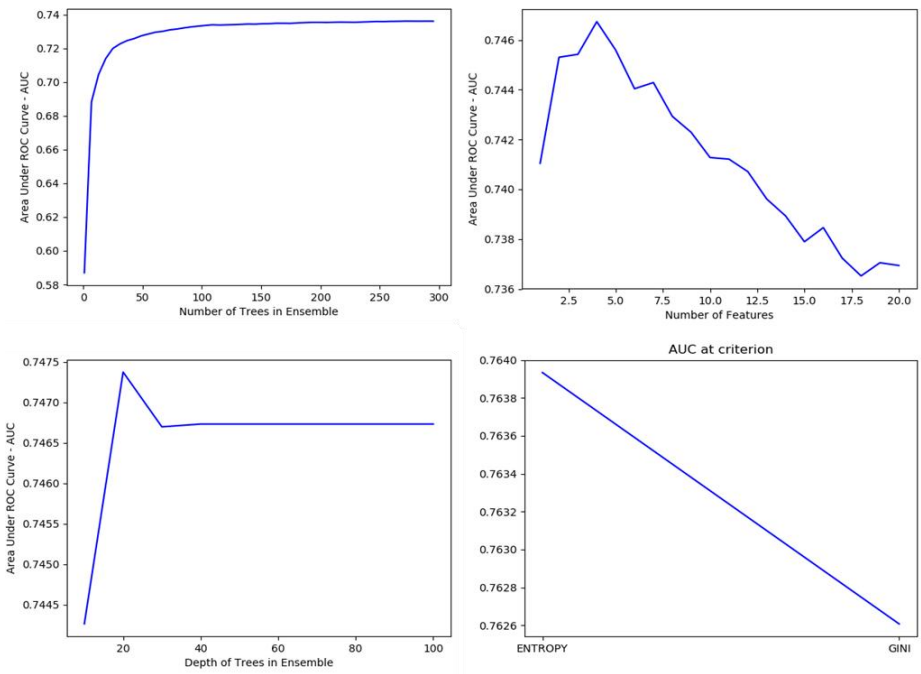


Figure 3 shows the impact of different parameter values on AUC during the process of generating a Random Forest. Among them, the upper left figure shows the trend of AUC as the number of decision trees continues to increase. The simulation of this process is achieved by adjusting the parameter ***n_estimators***. From the graph, it can be seen that when the number of base learners reaches 100, the overall performance of the model stabilizes basically without changing other conditions. The upper right figure shows the variation of AUC with ***max_features*** under the premise of having 200 base learners. It can be seen that when the maximum number of random features selected for splitting nodes is 4, the AUC value is the highest. The left figure shows how the maximum depth ***max_depth*** value of the base learner (decision tree) affects AUC. From the graph, it can be concluded that in order to save model training time and reduce model complexity, the maximum depth of the decision tree can be set within a certain range, because beyond this range, the performance of the model will not improve. The following figure on the right shows the advantages and disadvantages of two node splitting strategies, "***gini***" and "***entropy***". Obviously, on the dataset tested in this article, the latter has slightly better performance.

Figure 3. Optimization process of Random Forest parameters

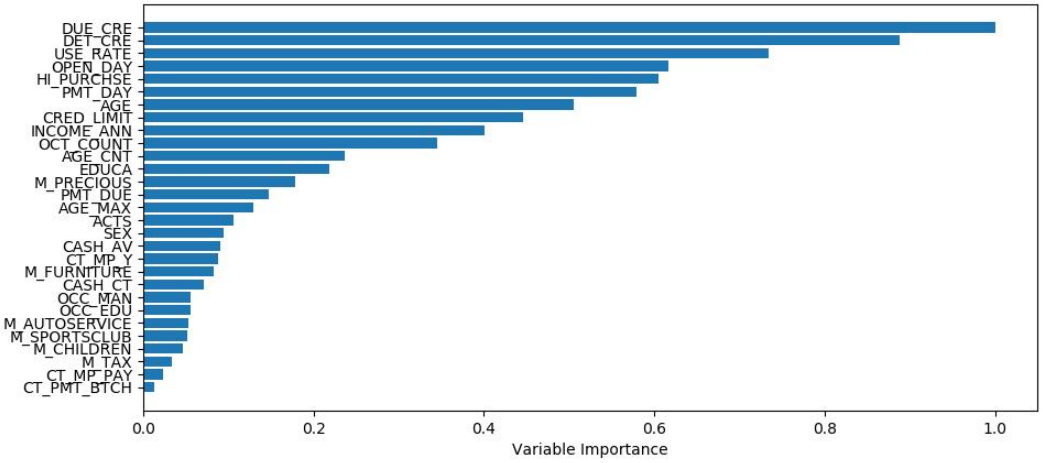


The initial training parameters for AdaBoost and Random Forest have been adjusted according to the above approach, respectively, and the final optimized output results were shown in Table 4.

Table 4. Improved model effect due to parameters adjustment

	AdaBoost		Random Forest	
	AUC	Precision	AUC	Precision
Default parameters	74.60%	67.31%	71.04%	64.56%
Adjusted parameters	74.90%	67.84%	76.39%	67.74%
Enhancement	0.41%	0.78%	7.54%	4.94%

Figure 4. Variable Importance



Based on contribution to decision-making process, the study outputs a variable importance ranking chart (Figure 4). By measuring the importance of feature variables, we can divided them into four categories.

Highlighted features (>0.7), which includes: overdue amount to limit ratio, maximum debt to limit ratio, and limit use rate

Important features (0.3-0.7), which includes: account opening duration, historical highest consumption amount, days from last repayment to billing date, age, credit limit, annual income, etc

Secondary features (0.1-0.3), which includes: number of overdue payments, education level, consumption frequency of key category merchants in the past year

Insensitive features (<0.1), which includes: gender, industry category, historical cash withdrawal, and historical installment

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